Design of pinhole sunglasses: aperture shape, lattice arrangement, ghost suppression and neural-network-accelerated inverse design

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Abstract

Pinhole sunglasses, opaque films perforated with a regular array of sub-millimetre apertures, attenuate light and sharpen the retinal image of uncorrected myopes, but their design couples several competing requirements: shrinking the apertures suppresses defocus blur while aggravating diffraction, and the lattice pitch fixes the light throughput, the appearance of ghost images and the coverage of the visual field. We formulate this design problem quantitatively on a reduced-eye model (power $P = 60$ D, image-side focal length $f' = 22.2$ mm, $\lambda = 550$ nm, film–cornea distance $L = 15$ mm) with a Scheiner-disc ghost criterion, $\Delta x = sM/P$ resolved when $\Delta x > \max(c_{\mathrm{tot}}, c_{\mathrm{min}})$, where the perceptual fusion floor $c_{\mathrm{min}} = 55$ ţm is calibrated to the clinical data of Miller and Johnson. A unified Monte-Carlo evaluator then maps the full design space of aperture shapes and lattice arrangements at fixed fill factor. Circular apertures are optimal: at fixed area the circle minimizes the circumscribed diameter, which gives it the highest acuity and the lowest ghost rate among the genuinely non-circular shapes in all tested conditions (tied only with the degenerate $\rho = 1$ ellipse), while elliptical apertures are quantitatively ruled out (up to $-44\,\%$ acuity and a ninefold ghost-rate increase) and annular apertures pay a sidelobe penalty of 31.5–62.5 % of the transmitted energy. Among eight lattice families the hexagonal lattice leads the hard-condition ranking (best score 0.693, versus 0.668 for square), the Vogel spiral dilutes ghost severity without reducing the ghost rate, and ring lattices are unsuitable for central-vision priority (on-axis ghost rate 1.0). The calibrated criterion opens a triple-win window: for $M = 1$–4 D we give designs with $d = 456$–720 ţm, $s = 824$–1352 ţm, $\eta = 0.18$–0.25, fused nearest-neighbour images ($g_{\mathrm{Sch}} \leq 1$) and full pupil coverage. A neural surrogate of the array evaluator (validation $R^2 = 0.979$–0.995) accelerates inverse design: gradient ascent through the frozen surrogate finds each optimum in 3.7 s (4.2× faster than direct random search at statistically equal quality, prediction–verification bias ≤ 0.003 in seven of eight conditions), and a trained inverse network returns a design in 0.5 ms ($\sim 10^4$–$10^5\times$ speed-up), with the one-off training cost recovered after about 24 design conditions. We close with an engineering specification: a hexagonal array of 530 ţm circular pinholes at 1.10 mm pitch ($\eta \approx 0.21$, about 1200 apertures on a 40 mm lens), compatible with EN ISO 12312-1 Category 2 and 3 sunglass transmission classes for substrate transmittances above 0.85 and in 0.38–0.85, respectively.

1 Introduction

The stenopeic effect, the sharpening of a defocused retinal image by a small aperture, is among the oldest known optical phenomena and remains a standard clinical tool: a pinhole occluder that improves a patient's visual acuity signals a refractive rather than pathological origin of reduced vision [4]. Pinhole *glasses*, which carry an array of apertures in an opaque sheet and thereby combine this depth-of-focus extension with a sunglass-like attenuation of light, have been marketed for decades, frequently accompanied by unsupported claims of vision training or myopia prevention; controlled evidence for such claims is absent, whereas the underlying

geometrical and diffractive optics is well understood and readily demonstrated in the teaching laboratory [3]. The problem has recently regained attention as an open design challenge, for example as an "invent yourself" problem of the International Young Physicists' Tournament (IYPT 2027), which asks precisely *how* the aperture diameter, lattice pitch and arrangement should be chosen to optimize a pinhole-sunglass device.

The single-pinhole trade-off is classical. A smaller aperture reduces the geometric (defocus) blur in proportion to its diameter but enlarges the diffraction (Airy) blur in inverse proportion, so an optimal diameter exists at which the two contributions balance [1,2,7]. For a focused eye viewing a distant object this optimum lies near $d_{\mathrm{opt}} = \sqrt{2.44\lambda f} \approx 172$ µm for visible light and the eye's focal length [1]; related criteria have been derived from self-imaging considerations [2,7] and, for a defocused single pinhole in front of the eye, from a full optical-transfer-function (OTF) analysis [5]. Turning a single pinhole into an *array* introduces further, competing constraints: the pitch relative to the aperture size fixes the fill factor (and hence the shading), pitches smaller than the pupil produce multiple overlapping retinal images (ghosts or, beyond a threshold, monocular diplopia) [3, 4], while an overly sparse array leaves dead zones in the visual field. Recent work on pinhole-array Maxwellian-view displays formalizes closely related geometric constraints for a comparable eye relief of $L = 15$ mm [6].

Three gaps remain in this picture. First, the ghost constraint has no established quantitative form: the naive resolvability criterion predicts diplopia at arbitrarily small refractive error, in contradiction with the clinical observation that multi-beam apertures produce Scheiner-disc monocular diplopia only from about 4 D upward [4]. Second, the aperture shape and the lattice arrangement have been treated anecdotally (commercial products use a square grid of about 3 mm pitch [3]), without a systematic comparison under a common fill-factor budget. Third, the coupled multi-objective design space is tedious to navigate by hand, and Monte-Carlo evaluation of array geometries, although cheap per sample, becomes expensive when repeated across conditions; whether neural surrogates can accelerate the search without sacrificing fidelity has not been quantified for this problem.

In this paper we (i) formulate a transparent, quantitative model of pinhole sunglasses on the basis of a reduced eye, combining defocus and diffraction blur, fill factor, a calibrated Scheiner-disc ghost criterion and pupil coverage into a multi-objective score (Sec. 2); (ii) explore the design space of aperture shapes (circle, square, diamond, ellipse, annulus) and lattice arrangements (square, hexagonal, rhombic, rings, Vogel spiral, jittered, apodized, two-scale) with a unified Monte-Carlo evaluator at fixed fill factor (Sec. 3); (iii) train a neural-network surrogate of the array evaluator and quantify two neural inverse-design routes, gradient ascent through the frozen surrogate and a direct inverse-design network, against a random-search baseline (Sec. 4); (iv) report score-optimal and fusion-compatible triple-win designs for myopia of 1–4 D (Sec. 5); and (v) condense the findings into an engineering specification, including film-thickness field-of-view checks, sunglass-standard transmission classes and fabrication tolerances (Sec. 6). Limitations are discussed in Sec. 7.

2 Physical model

2.1 Single pinhole: defocus–diffraction trade-off

We adopt the reduced-eye model in which the eye is represented by a single refracting surface of power $P = 60$ D and image-side focal length $f' = n/P = 1.333/60 \approx 22.2$ mm, at a wavelength $\lambda = 550$ nm representative of photopic vision; the same 22–23 mm scale for the eye's image-side focal length is used in Refs. [1,3]. A myopic eye with refractive error M (in dioptres, $M > 0$) viewing a distant object through a pinhole of diameter d produces a geometric blur circle on the retina whose diameter scales linearly with the aperture and the refractive error [3,4],

$$c_{\mathrm{geom}} = d\,\frac{M}{P}. \tag{1}$$

Diffraction at the aperture contributes the Airy-disc diameter [1, 3]

$$c_{\mathrm{diff}} = \frac{1.22\,\lambda f'}{d}. \tag{2}$$

Treating the two contributions as independent, we combine them in quadrature (root-sum-square, RSS),

$$c_{\mathrm{tot}} = \sqrt{c_{\mathrm{geom}}^2 + c_{\mathrm{diff}}^2}. \tag{3}$$

A normalized visual acuity is defined relative to the diffraction limit of the emmetropic eye at the pupil diameter under consideration,

$$a = \frac{c_{\mathrm{ref}}}{c_{\mathrm{tot}}}, \qquad c_{\mathrm{ref}} = \frac{1.22\,\lambda f'}{p}, \tag{4}$$

with p the pupil diameter of the condition being evaluated (so that the $p = 4$ mm conditions are normalized against a twice stricter reference than the $p = 2$ mm conditions); $a = 1$ corresponds to the emmetropic diffraction limit, and for optimization a is clipped to $[0, 3]$ and rescaled, $a_{\mathrm{norm}} = \min(a, 3)/3$.

The trade-off embodied in Eq. (3) is visualized in Fig. 1. Minimizing c_{tot} with respect to d yields the analytic optimum

$$d_{\mathrm{opt}}^2 = \frac{P \cdot 1.22\,\lambda f'}{M}, \tag{5}$$

which reduces, in the limit of vanishing defocus, to the classical pinhole-camera criterion of Young, $d_{\mathrm{opt}} = \sqrt{2.44\lambda f'} \approx 172$ µm at $f' = 22.2$ mm [1]; the closely related self-imaging criterion $d_{\mathrm{opt}} \approx 1.9\sqrt{\lambda z}$ of Newman and Rible (quoted by Bryngdahl [2, 7]) gives ≈ 209 µm for $z = f'$, and the OTF-based analysis of Ref. [5] finds optima of order 0.2 mm for a focused eye. With a finite myopic defocus the optimum shifts to larger diameters: for $M = 3$ D, Eq. (5) predicts $d_{\mathrm{opt}} \approx 546$ µm.

2.2 Pinhole array: fill factor, ghosts and coverage

The geometry is summarized in Fig. 2. An opaque film carrying a regular array of pinholes of diameter d and pitch s is placed at an eye relief $L = 15$ mm in front of the cornea, matching both typical eyewear and the geometry of Ref. [6]. For square and hexagonal lattices the fill factor (the open fraction, i.e. the photopic transmission relative to the unobstructed pupil) is purely geometrical,

$$\eta_{\mathrm{sq}} = \frac{\pi d^2}{4s^2}, \qquad \eta_{\mathrm{hex}} = \frac{\pi d^2}{2\sqrt{3}\,s^2}. \tag{6}$$

Commercial pinhole glasses use a square grid with a pitch of about 3 mm [3]; we target the darker range typical of sunglasses, $\eta \in [0.05, 0.25]$, with a preference for $\eta \approx 0.15$.

Each pinhole projects its own image of the scene onto the retina. When the pitch is smaller than the pupil, two beams selected by adjacent pinholes can enter the pupil simultaneously and form two superimposed images [3]. For a distant object the two beams are parallel and are separated by s at the pupil; the myopic eye casts each of them into a blur circle, and the centres of the two circles are separated by

$$\Delta x = s\,\frac{M}{P}, \tag{7}$$

which is the exact paraxial result for lens-induced defocus (we verified it against a reduced-eye matrix-optics derivation and numerical ray tracing); the alternative axial-defocus convention $sM/(P + M)$ differs by at most 9 % for $M \le 6$ D and 6 % for $M \le 4$ D. Because both Δx and c_{geom} are proportional to M, a resolvability criterion based on $\Delta x > c_{\mathrm{tot}}$ alone has no threshold structure in M and would predict diplopia at arbitrarily small refractive error, in

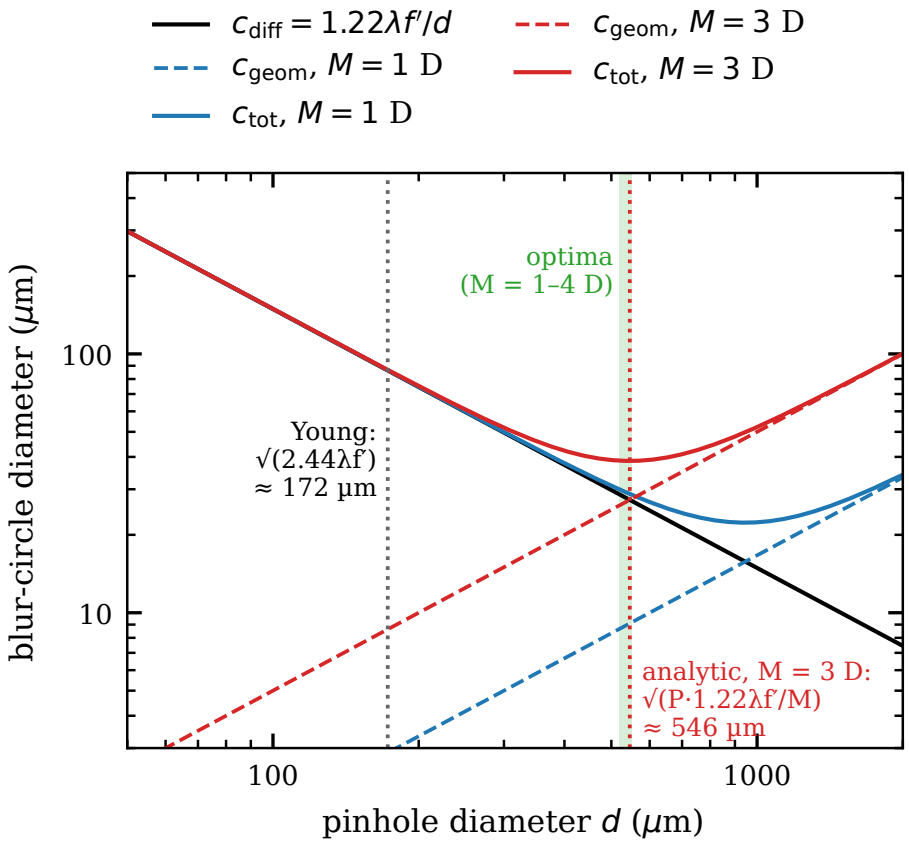


Figure 1: Blur-circle diameter on the retina as a function of pinhole diameter d (log–log scale): diffraction blur c_{diff} [Eq. (2)], geometric (defocus) blur c_{geom} [Eq. (1)] and RSS-combined total blur c_{tot} [Eq. (3)] for myopia of $M = 1$ and 3 D ($P = 60$ D, $f' = 22.2$ mm, $\lambda = 550$ nm). Dotted vertical lines mark the zero-defocus Young criterion $d_{\text{opt}} = \sqrt{2.44\lambda f'} \approx 172$ µm and the analytic defocus optimum of Eq. (5) for $M = 3$ D (546 µm); the shaded band indicates the range of optimized diameters found for $M = 1$–4 D (Sec. 5).

contradiction with the clinical observation that multi-beam apertures produce Scheiner-disc monocular diplopia only from about 4 D upward [4]. We therefore add an absolute perceptual fusion floor $c_{\min}$ and define the Scheiner ghost metric

$$g_{\text{Sch}} = w(s,d)\,\frac{\Delta x}{\max(c_{\text{tot}},\, c_{\min})}, \qquad w = \text{clip}\left(\frac{p+d-s}{d},\, 0,\, 1\right), \tag{8}$$

where the smooth gating factor w expresses that the two beams can overlap the pupil simultaneously only for $s < p + d$ [3]. Calibrating $c_{\min}$ against the multi-beam geometry of Ref. [4] ($s \approx 3$ mm, $d \approx 1.5$ mm, dilated pupil $p = 8$ mm) so that $M = 1$ D fuses ($g_{\text{Sch}} = 0.91$) while $M = 4$ D is resolved ($g_{\text{Sch}} = 1.99$) yields $c_{\min} \approx 55$ µm (≈ 8.5 arcmin at the retina), to be read as a perceptual blur-tolerance floor rather than a diffraction or cone-sampling resolution limit; the admissible calibration interval is $[50, 200)$ µm. Note that the separation (7) is independent of the eye relief L (parallel beams are invariant under transverse translation), so within this mechanism moving the film closer to the cornea does not by itself suppress ghosts.

For comparison we also retain the eye-rotation displacement model, in which the eye fixates the same object point through adjacent pinholes and the two retinal images are displaced by sf'/L; the corresponding metric

$$g = \frac{sf'/L}{c_{\text{tot}}} \tag{9}$$

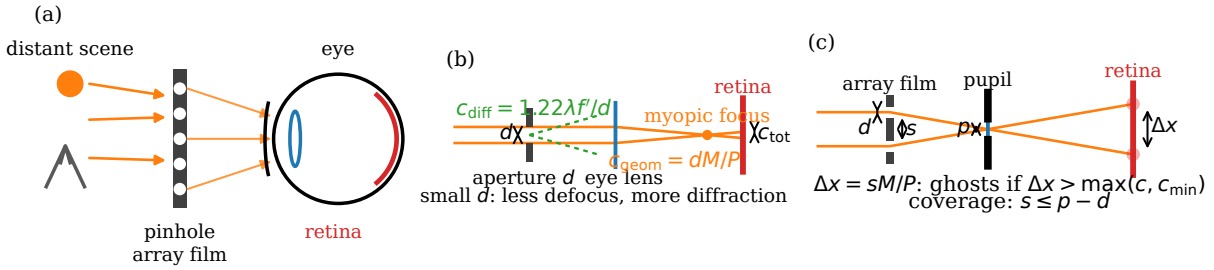


Figure 2: The pinhole-sunglass design problem (schematics, not to scale). (a) Device concept: light from a distant scene passes an opaque film perforated with a pinhole array mounted in front of the eye, combining sunglass-like attenuation with the stenopeic sharpening of the defocused retinal image. (b) Single-aperture physics: in a myopic eye the rays admitted by a pinhole of diameter d focus in front of the retina, leaving a geometric blur $c_{\text{geom}} = dM/P$ [Eq. (1)], while diffraction at the aperture contributes $c_{\text{diff}} = 1.22\lambda f'/d$ [Eq. (2)]; shrinking d reduces the former but enlarges the latter. (c) Array geometry: adjacent pinholes (pitch s) admit simultaneous beams whose blur-circle centres are separated by $\Delta x = sM/P$ [Eq. (7)], perceived as ghosts when $\Delta x > \max(c_{\text{tot}}, c_{\min})$ [Eq. (8)], and a pupil of diameter p keeps at least one pinhole in view only if $s \le p - d$ [Eq. (10)].

exceeds unity by one to two orders of magnitude for feasible designs at $L = 15$ mm ($g \approx 40$–66 at the optima of Sec. 5) and therefore cannot discriminate between designs. It is this saturating metric that enters the composite score of Sec. 2.4; all quantitative ghost conclusions below are drawn from the calibrated metric (8) or from the Monte-Carlo evaluator of Sec. 2.4. A structurally similar metric, stripe widths proportional to pitch times a distance ratio, appears in the analysis of pinhole-array displays [6].

Finally, as the eye rotates, the pupil of diameter p must always find at least one pinhole in front of it; otherwise dead zones appear in the visual field [3,6]. We quantify this by the coverage

$$\text{cov} = \min\left(1, \max\left(0, \frac{p - d}{s}\right)\right), \tag{10}$$

which equals unity when $s \le p - d$. Physically meaningful designs must also respect the non-overlap constraint $s \ge 1.2\,d$.

2.3 Aperture-shape corrections to the diffraction term

For non-circular apertures the diffraction term (2) changes its coefficient and, for anisotropic apertures, becomes direction dependent. A square aperture of side a_{sq} produces a sinc^2 point-spread function whose first zero lies at $\lambda f'/a_{\text{sq}}$, corresponding to a coefficient 1.0 in place of 1.22; an elliptical aperture with semi-axes (a_e, b_e) diffracts anisotropically, with the narrower axis producing the wider diffraction spread. In the evaluator the diffraction and geometric-blur terms are implemented per shape as follows. For circular and annular apertures Eq. (2) is evaluated at the circumscribed diameter d_{circ} of the equal-area circle (outer diameter for the annulus), and the geometric blur (1) uses the same d_{circ}. Square and diamond apertures use the exact sinc^2 coefficient 1.0 with the side length, $c_{\text{diff}} = \lambda f'/a_{\text{sq}}$, and the geometric blur is taken along the circumscribed diameter. Elliptical apertures are treated anisotropically axis by axis, $c_{\text{tot}} = \sqrt{(2a_eM/P)^2 + (1.22\lambda f'/2a_e)^2}$ per axis; the acuity uses the most unfavourable (larger) axis, while the Monte-Carlo pair test uses the geometric mean of the two axes. For an annular aperture with inner-to-outer radius ratio w_r the outer-diameter approximation is calibrated against the exact Fraunhofer integral: the true first-zero coefficient is 1.115 ($w_r = 0.3$) and 0.947 ($w_r = 0.6$), so the approximation overestimates c_{diff} by factors 1.09 and 1.29. The error is immaterial for the reported metrics, because $c_{\text{diff}} \approx 9$–$11$ ţm is much smaller than

both c_{geom} (22–120 ţm) and $c_{\min} = 55$ ţm, and correcting it changes c_{tot} by less than 2 %. It is nevertheless systematically optimistic about annular apertures, because the exact pattern places 31.5 % ($w_r = 0.3$) or 62.5 % ($w_r = 0.6$) of the transmitted energy outside the main lobe (circular aperture: 16 %), energy that appears as veiling glare rather than in any of our metrics (Sec. 3.2).

2.4 Multi-objective score and Monte-Carlo array evaluation

For the closed-form model, the four criteria are combined into a single scalar score,

$$S = w_a\, a_{\mathrm{norm}} + w_\eta\, \mu_\eta + w_g\, \mu_g + w_c\, \mathrm{cov}, \tag{11}$$

with weights $(w_a, w_\eta, w_g, w_c) = (1.0, 0.8, 1.0, 0.6)$. The fill-factor and ghost criteria enter through smooth membership functions,

$$\mu_\eta = \exp\left[-\frac{(\eta - 0.15)^2}{2(0.08)^2}\right], \qquad \mu_g = \sigma\left(2\left(1 - \frac{g}{g_{\max}}\right)\right), \tag{12}$$

where $\sigma(z) = 1/(1+e^{-z})$ and $g_{\max} = 1$ is the assumed image-fusion threshold. The ghost metric entering μ_g is the eye-rotation metric g of Eq. (9); as shown in Sec. 5 it saturates for all feasible designs and hence does not influence the optima, which are re-examined with the calibrated Scheiner metric in Sec. 5.2.

Array geometries beyond the two periodic lattices, and the interplay of ghosts with dead zones, require a statistical treatment. We therefore evaluate every array design by Monte-Carlo simulation: object directions are drawn uniformly in solid angle within a $\pm 10^\circ$ field, the pupil (diameter p, fixed at the array centre or wandering within the unit cell of a periodic lattice, or within the design's wander disc for the non-periodic families) samples the beams admitted through all apertures, and each pair of simultaneously admitted beams is tested for resolvability with the criterion $\Delta x_{ij} > \max(\bar{c}_{\mathrm{tot}}, c_{\min})$, where $\Delta x_{ij} = s_{ij} M/P$ uses the actual pair separation s_{ij} and $\bar{c}_{\mathrm{tot}}$ is the mean total blur of the two apertures (relevant only for the variable-aperture families). The evaluator returns the ghost rate (fraction of trials with at least one resolvable pair), the blind rate (fraction of trials with no admitted beam), the pair-resolved fraction (the fraction of admitted pairs that are resolvable, a measure of ghost severity), and the acuity; the score (11) is then computed with the ghost membership applied to the ghost rate and the coverage replaced by $1 -$ (blind rate). In the Monte-Carlo evaluator the score is normalized by the weight sum $\sum w_i = 3.4$, so that $S \in [0, 1]$; the closed-form tables of Sec. 5 instead use the unnormalized form of Eq. (11) ($S \leq 3.4$). All array studies below use circular apertures unless stated otherwise, normalize every design to $\eta = 0.15$ exactly by construction, and use 1000–5000 trials per configuration (binomial standard error $\sigma \leq 0.016$ at 1000 trials). One evaluation costs 18.2 ms on average at 1000 trials (26 ms at 2000 trials), which motivates the surrogate study of Sec. 4.

3 Design space exploration: shapes and lattices

3.1 Lattice arrangements

We compared eight arrangement families, square, hexagonal (hex), rhombic, concentric rings, Vogel spiral, randomly jittered, apodized (Gaussian aperture-size taper) and two-scale (checkerboard of two aperture sizes), on the unified evaluator. The rhombic sweep continuously interpolates between hex ($\theta = 60^\circ$) and square ($\theta = 90^\circ$) and tests the covering-radius theory: a rhombic lattice with primitive vectors of length s and angle θ has a largest empty-circle radius

$$R_{\mathrm{cov}} = \frac{s}{2\cos(\theta/2)}, \tag{13}$$

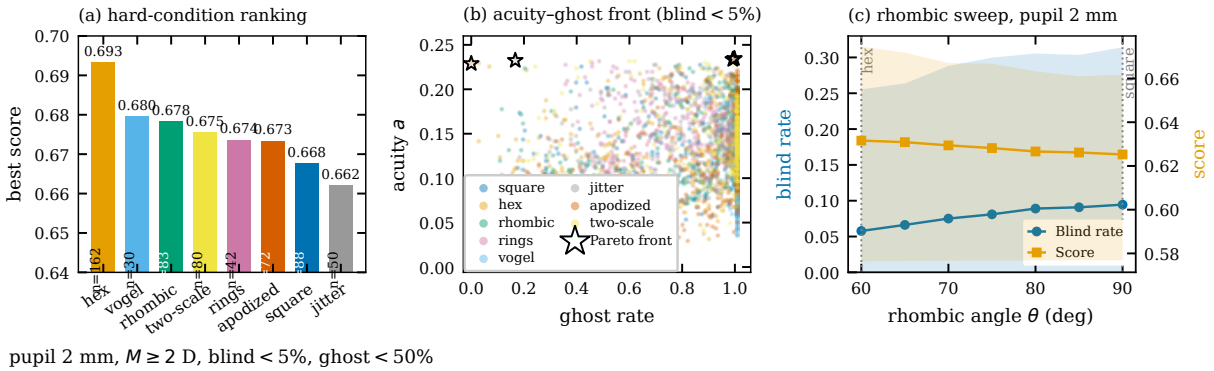


Figure 3: Lattice-arrangement comparison (unified Monte-Carlo evaluator, $\eta = 0.15$ for all designs). (a) Best score per arrangement family under the hard condition (pupil 2 mm, $M \geq 2$ D, blind rate $< 5\,\%$, ghost rate $< 50\,\%$); bar labels give the best score and the number n of designs passing the filter (14 040-design dataset). Hexagonal leads at 0.693. (b) Acuity versus ghost rate for all designs with blind rate $< 5\,\%$ (pupil 2 mm, $M \geq 2$ D); stars mark the four non-dominated points (one hex, one rings, two apodized) of the acuity–ghost front. (c) Rhombic sweep at $M = 3$ D, pupil 2 mm: blind rate (blue, left axis) and score (orange, right axis) versus rhombic angle θ, mean and range over seven pitches $s = 1$–4 mm; the blind rate rises monotonically (within Monte-Carlo noise) from the hex ($\theta = 60°$) to the square ($\theta = 90°$) limit, confirming the covering-radius theory [Eq. (13)].

which is minimized at $\theta = 60°$ ($s/\sqrt{3}$, hex) and maximized at $\theta = 90°$ ($s/\sqrt{2}$, square); blind zones onset when R_{cov} exceeds the capture radius $(p + d_{\mathrm{eff}})/2$. The measurements confirm the theory quantitatively [Fig. 3(c)]: at $M = 3$ D and $p = 2$ mm the blind rate rises monotonically with θ within the Monte-Carlo noise (all 42 pitch-resolved adjacent steps up to noise-level fluctuations of at most 0.004, mean 0.058 at hex to 0.095 at square), the measured blind-zone onset tracks the theoretical θ^* within one grid step, and the ghost rate is essentially θ-independent (0.417 hex versus 0.409 square) because it is governed by the nearest-neighbour distance s. The score therefore increases monotonically (within noise) towards the hex end (0.625 → 0.632), and no intermediate angle beats hexagonal packing.

Figure 3(a) ranks the eight families by their best score under a hard condition (pupil 2 mm, $M \geq 2$ D, blind rate $< 5\,\%$, ghost rate $< 50\,\%$): hex leads at 0.693 (with 162 designs in the high-quality region, the largest contingent), followed by Vogel (0.680), rhombic (0.678), two-scale (0.675), rings (0.674), apodized (0.673), square (0.668) and jittered (0.662). Two caveats accompany this ranking. First, the unconstrained (blind, ghost) two-objective front is degenerate: the fusion floor c_{min} makes the $(0, 0)$ corner genuinely reachable (28 designs), but those designs have a median acuity of only 0.06, so the informative trade-off is acuity versus ghost rate at bounded blind rate, whose front contains four designs: one each of hex and rings and two of the apodized family [Fig. 3(b)]. Second, the score distribution is double-banded in M (a ghost-free upper band ≈ 0.63 and a ghost-active lower band ≈ 0.57), within which the family ordering is stable.

Three non-periodic families deserve specific comment. The Vogel spiral was motivated by the hypothesis that aperiodic pair distances suppress coherent ghosts. The data split this hypothesis in two: the ghost *rate* is not reduced but slightly increased (in an equal-density 98-point grid the Vogel ghost rate exceeded that of hex by more than 0.02, $\approx 2.5\sigma$ of the Monte-Carlo noise, at 38 points and undercut it at only 3; dataset median 1.000, the worst of all families), because the ghost rate is a binary quantity dominated by the shortest pair, and the Vogel nearest-neighbour distribution has a short tail (median $\approx 0.88\, s_{\mathrm{hex}}$). The pair-resolved fraction, by contrast, is genuinely reduced (median 0.52, while all other families sit at 0.93–1.00; transition-zone reductions reach 1.000 → 0.709 at $p = 2$ mm and 1.000 → 0.341 at $p = 4$ mm): the hex

pair distances are discretely repeated, so once the resolvability threshold crosses s, all nearest-neighbour pairs become resolvable together and superimpose as a single strong double image, whereas the Vogel distances spread the resolved ghosts into multiple weak images. The spiral dilutes ghost severity without preventing ghosts, an advantage outside the score's ghost-rate definition. Ring lattices fail the central-vision test: with the pupil centred on the array, the on-axis ghost rate is 1.000 (the first ring sits inside the capture disc, so the whole ring enters the pupil), against 0.000 for an equal-density hex array aligned on a single aperture; the hex value is an ideal-alignment upper bound, since real fixation wanders. If central-vision priority is desired, the apodized taper (largest aperture on axis) is the appropriate structure, not rings. The apodized family is a genuine trade-off arrangement: it reduces large-pitch blind zones (blind rate $0.115 \to 0.05$ at $s = 3$ mm as the taper α grows to 2) at the price of a higher transition-region ghost rate ($0.62 \to 0.72$ at $s = 2$ mm, $M = 3$ D, taper $\alpha = 1$; 0.79 at $\alpha = 2$) and a reduced acuity of the reference aperture; it reaches the highest single score in the dataset (0.894 at $M \approx 0$) but does not beat hex under hard conditions. The two-scale family is indistinguishable from its square baseline (Δscore within $[-0.008, +0.013]$, at the Monte-Carlo noise edge) and adds no Pareto gain.

3.2 Aperture shapes

Nine shape configurations (circle; ellipse with axis ratio $\rho = 1.0, 0.7, 0.5, 0.3$; square; diamond; annulus with $w_r = 0.3, 0.6$) were compared on hexagonal lattices at exactly $\eta = 0.15$ across four conditions ($M \in \{1, 3\}$ D $\times$ pupil $\{2, 4\}$ mm, 5000 trials per cell). At fixed area and pitch, a shape's only degree of freedom is its circumscribed diameter

$$d_{\mathrm{circ}} = \kappa\sqrt{A}, \tag{14}$$

with $\kappa = 1.128$ (circle, the isoperimetric minimum), 1.414 (square, diamond), $1.128/\sqrt{\rho}$ (ellipse) and $1.128/\sqrt{1 - w_r^2}$ (annulus). The circumscribed diameter controls three quantities at once: the geometric blur $c_{\mathrm{geom}} = d_{\mathrm{circ}} M/P$ (hence the acuity), the capture radius $(p + d_{\mathrm{circ}})/2$ (hence the blind rate) and the number of simultaneously admitted beam pairs (hence the ghost rate). Shape selection is therefore a three-way trade-off, and the circle, as the isoperimetric minimizer, is expected to excel at acuity and ghosts while paying a small blind-rate penalty.

The measurements (Fig. 4) confirm this picture with two robust findings. First, the circle has the highest acuity and the lowest ghost rate among the genuinely non-circular shapes in all four conditions, independent of pitch (tied with the degenerate $\rho = 1$ ellipse; every difference from a non-circular shape exceeds the Monte-Carlo noise); its cost is a moderately higher blind rate (4.8 % versus 0 % for the square at $M = 1$ D, $p = 2$ mm), which pitch tuning can recover. The annulus with $w_r = 0.3$ is the closest substitute (d_{circ} only 4.8 % larger), with square and diamond in between. Second, the differences are genuinely small among all compact shapes: score gaps stay below ≈ 0.013 and the ordering reshuffles under a ± 15 % pitch perturbation, so the shape ranking beyond the circle's acuity and ghost advantages is not robust. A direct manifestation of the fusion floor is visible at $M = 1$ D and $p = 2$ mm, where $\Delta x_{\mathrm{nn}} = sM/P = 53.3$ ţm$< c_{\min}$ and all nine shapes have ghost rate exactly zero: below the floor, shape is irrelevant to ghosts. (At $p = 4$ mm the pitch is larger, $s = 6.0$ mm, the separation 100 ţm exceeds the floor, and ghost rates of 0.06–0.63 reappear.)

The ellipse deserves a quantitative rebuttal, because the intuitive proposal (keep the short axis sharp, compress ghosts along the long axis) fails on three grounds. (i) The Scheiner separation $\Delta x = s_{ij} M/P$ is an isotropic geometric quantity that depends only on the pair distance, and a hexagonal lattice has nearest-neighbour pairs in three directions, so suppressing ghosts along one axis leaves the others unresolved; covering all directions with anisotropic blur is equivalent to isotropically enlarging the blur. (ii) At low M the criterion is locked by $c_{\min}$ and insensitive to any blur change: reducing ρ from 1.0 to 0.3 leaves the ghost rate at zero while the acuity drops from

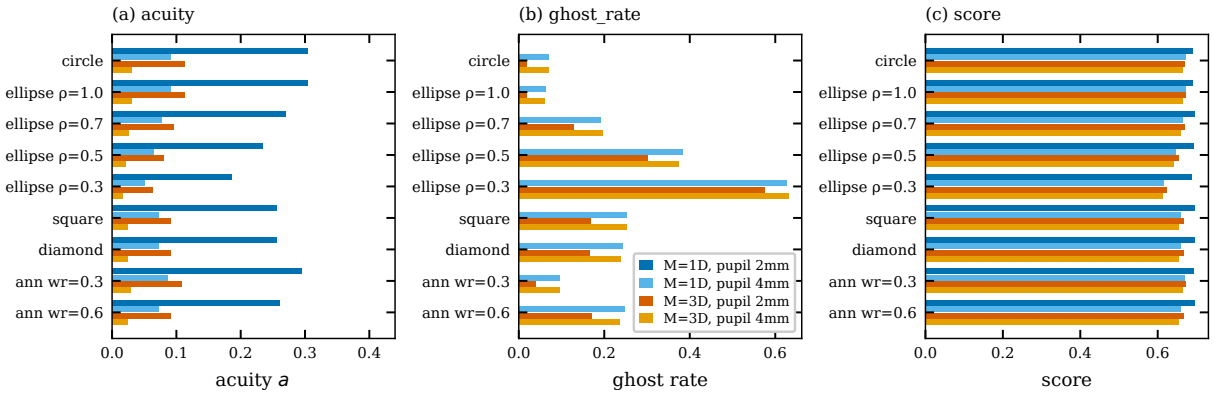


Figure 4: Aperture-shape comparison on hexagonal lattices at fixed fill factor $\eta = 0.15$ (5000 Monte-Carlo trials per cell; with a common seed the ellipse $\rho = 1.0$ reproduces the circle exactly, as verified, while the plotted cells use independent seeds and differ by Monte-Carlo noise). (a) Normalized acuity; (b) ghost rate; (c) composite score. The circle has the highest acuity and the lowest ghost rate among the genuinely non-circular shapes in all four conditions (tied with the degenerate $\rho = 1$ ellipse; all differences from non-circular shapes exceed the Monte-Carlo noise). At $M = 1$ D and $p = 2$ mm all ghost rates vanish because the nearest-neighbour separation $\Delta x_{\mathrm{nn}} = 53.3$ ţm lies below the fusion floor $c_{\mathrm{min}} = 55$ ţm; at $p = 4$ mm ($s = 6.0$ mm) the separation (100 ţm) exceeds the floor and ghosts reappear. Score differences among the compact shapes are $\lesssim 0.013$ and reshuffle with pitch, so only the acuity and ghost-rate advantages are robust.

0.304 to 0.186 (−39 %). (iii) The required magnitudes are unreachable: fusing nearest neighbours at $M = 3$ D needs $c_{\mathrm{tot}} \geq 160$ ţm, which geometrically requires a long axis $a \geq s = 3.2$ mm ($\rho = 0.165$ at fixed η) and diffractively a short axis $b \leq 1.22\lambda f' P/(sM) \approx 0.093$ mm, physically absurd at fixed fill factor. Measured at $\rho = 0.3$, the long-axis blur (119 ţm) still falls short of 160 ţm: the full account from $\rho = 1.0$ to 0.3 at $M = 3$ D, $p = 4$ mm is acuity 0.030 → 0.017 (−44 %), ghost rate 0.070 → 0.630 (ninefold worse, because the enlarged circumscribed disc admits more pairs) and score 0.666 → 0.612. The annulus, finally, carries a cost that the evaluator does not even register: the exact Fraunhofer pattern places 31.5 % ($w_r = 0.3$) or 62.5 % ($w_r = 0.6$) of the energy in sidelobe rings (circle: 16 %), appearing as veiling glare, and at fixed η the outer diameter grows by 25 % at $w_r = 0.6$, so the measured behaviour coincides with that of equal-circumscribed-diameter square and diamond apertures. The narrow-mainlobe advantage of annular apertures is entirely cancelled by the geometric-blur increase, and the sidelobe penalty makes their real perceptual quality worse than the metrics suggest. We therefore adopt circular apertures for all device designs below.

4 Neural-network-accelerated inverse design

4.1 Surrogate of the array evaluator

The design space of Sec. 3 was sampled into a dataset of 14 040 designs (780 per legal shape–arrangement combination, M uniform in $[0, 6]$ D, pupil 2 or 4 mm, pitch s log-uniform and the remaining arrangement parameters uniform), each evaluated by the Monte-Carlo evaluator at 1000 trials. The surrogate is a multilayer perceptron $29 \to 512 \to 512 \to 512 \to 4$ (SiLU activations) mapping the design encoding (13 one-hot shape/arrangement indicators plus 16 continuous parameters) to ghost rate, blind rate, acuity and score. Rate labels use binary-cross-entropy soft targets, which tolerate the binomial label noise ($\sigma \leq 0.016$) and the $[0, 1]$ truncation; acuity is regressed in $\log(1+a)$ space. Training (Adam, cosine annealing, 800 epochs, 90/10 split) took 281 s on a CPU. Validation fidelity is summarized in Table 1: all outputs reach $R^2 \geq 0.979$,

Table 1: Fidelity of the array-evaluator surrogate on the held-out validation split ($n = 1404$). The score is reported both for the direct output head and composed from the other three heads through the exact $\eta \equiv 0.15$ identity.

Output	R^2	RMSE
ghost rate	0.9840	0.0536
blind rate	0.9913	0.0061
acuity ($\log(1+a)$)	0.9953	0.0064
score (direct head)	0.9793	0.0072
score (composed)	0.9847	0.0061

Table 2: Path A (gradient ascent through the frozen surrogate) on eight design conditions. Predicted and true (verified, 2000 Monte-Carlo trials) scores refer to the best candidate after re-evaluation of the top three. Optimization times are CPU wall clock.

M (D)	pupil (mm)	combination	pred. score	true score	dev.	time (s)
1	2	hex/ellipse	0.7059	0.7036	+0.0023	3.7
1	4	hex/ellipse	0.6875	0.6720	+0.0155	3.8
2	2	hex/ellipse	0.6929	0.6935	−0.0006	3.6
2	4	rhombic/circle	0.6568	0.6570	−0.0002	3.8
3	2	vogel/ellipse	0.6776	0.6770	+0.0006	3.7
3	4	two-scale/circle	0.6565	0.6556	+0.0009	3.6
4	2	apodized/annulus	0.6727	0.6723	+0.0003	3.6
4	4	two-scale/circle	0.6559	0.6550	+0.0009	3.6

the ghost-rate head being limited by the Monte-Carlo noise floor of the labels. Because every design in the dataset has $\eta \equiv 0.15$ exactly, the composite score is an exact linear combination of the other three heads (agreement with the labels to better than 10^{-7}); path A below optimizes this composed score, which is therefore strictly consistent with the true objective.

4.2 Path A: gradient ascent through the frozen surrogate

For each design condition (M, p) we maximize the surrogate score over the 18 legal shape–arrangement combinations: the combination one-hot is fixed in turn, only the applicable continuous parameters are optimized (sigmoid-projected back to the legal sampling domain, with a closed-form differentiable effective diameter reproducing the evaluator's η scaling to $\leq 4 \times 10^{-7}$ for 16 of 18 combinations), with 48 restarts and 300 Adam steps per combination. The top three candidates across combinations are then re-evaluated with the true Monte-Carlo evaluator (2000 trials), and the final design is chosen by true score. Table 2 lists the results for the eight conditions $M \in \{1, 2, 3, 4\}$ D $\times$ pupil $\{2, 4\}$ mm. The mean optimization time is 3.68 s per condition (gradient ascent only; the top-three re-evaluation adds ≈ 0.1 s), and the prediction–verification bias is +0.0025 on average: seven of the eight conditions verify within ± 0.003, while the $M = 1$ D, $p = 4$ mm condition overestimates by 0.016, a boundary case discussed in Sec. 4.5. Four of the selected "ellipse" solutions have axis ratios 0.994–0.999, i.e. they are effectively circular, consistent with Sec. 3.2; the single annulus solution ($M = 4$ D, $p = 2$ mm, on an apodized lattice) has $w_r \approx 0.89$, i.e. a thin ring.

4.3 Path B: inverse-design network

The second route trains a network $(M/6, (p-2)/2) \rightarrow$ (shape logits, arrangement logits, 12 sigmoid-bounded continuous parameters) to map a design condition directly to a full design in one forward pass. The teacher for each bin of conditions (bin centres aligned to integer M)

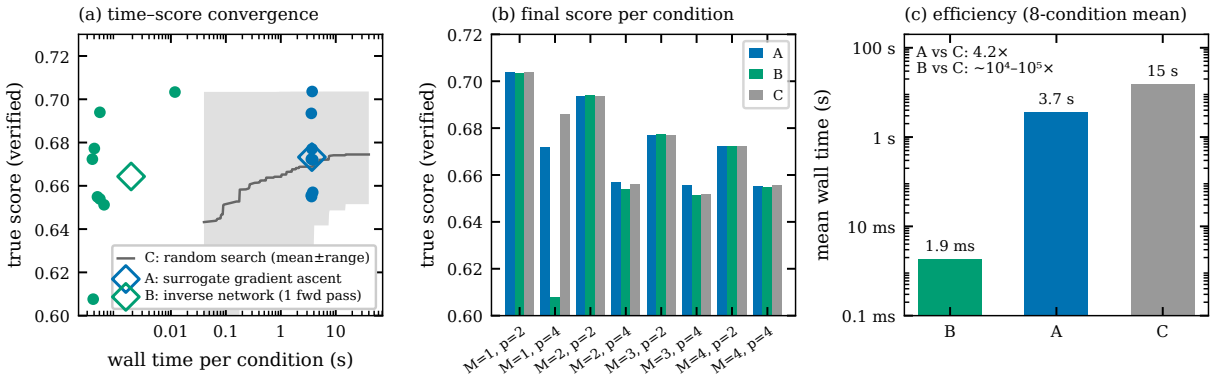


Figure 5: Efficiency benchmark of the neural inverse-design routes. (a) True-score convergence versus wall time per condition: the grey band and curve give the range and mean of path C (random search on the true evaluator) over the eight conditions; blue and green markers show the per-condition results of path A (surrogate gradient ascent) and path B (inverse network), with open diamonds at the means. (b) Final verified score per condition for the three routes. (c) Mean wall time per condition (log scale); annotations give the speed-ups relative to path C.

Table 3: Inverse-design benchmark over the eight conditions of Table 2: mean verified score, wall time per condition and speed-up relative to path C (random search, 600 true evaluations per condition).

Method	mean true score	time per condition	speed-up
A: frozen-surrogate gradient ascent	0.6733	3.68 s	4.2×
B: inverse-design network	0.6643	0.5 ms	$\sim 10^4$–$10^5\times$
C: random search (true evaluator)	0.6745	15.5 s	1×

is the dataset sample with the highest true score; training took 68 s. Evaluated on the eight conditions, the inverse network returns designs in ~ 0.5 ms whose verified scores average 0.6643: six of eight conditions tie with the random-search baseline within ± 0.001, of the remaining two the $M = 2$, $p = 4$ mm condition falls short by 0.002 and the $M = 1$, $p = 4$ mm condition by 0.08. The shortfall is structural: the teacher labels are single-sample Monte-Carlo maxima, so the inverse network can imitate but not exceed the data, and bin-boundary mismatch can blend two different teacher designs into a bad hybrid (one condition scored 0.562 before the bin alignment was fixed). Path B is therefore suited to seeding or batch screening, not to final designs.

4.4 Efficiency benchmark

Figure 5 and Table 3 compare the two neural routes against path C, random search over the same sampling distribution calling the true evaluator directly (600 evaluations per condition). Path A matches path C in mean true score (0.6733 versus 0.6745; four of eight conditions tied or better, the worst deficit -0.014 at the $M = 1$ D, $p = 4$ mm boundary condition of Sec. 4.5) in 3.68 s instead of 15.5 s, a 4.2× speed-up; at equal wall-time budget the advantage is larger, since path C reaches only ≈ 0.67 in 3.7 s [Fig. 5(a)]. Path B is 10^4–$10^5\times$ faster per condition but capped by its teacher labels. The one-off costs are 257 s for dataset generation and 281 s for surrogate training; at ~ 12 s saved per condition, the surrogate route breaks even after about 46 conditions (the training cost alone after about 24), and the advantage grows with the size of the condition sweep.

4.5 Applicability boundaries

Four boundaries delimit the neural workflow. First, the rate labels carry a Monte-Carlo noise floor ($\sigma \leq 0.016$ at 1000 trials, and 62 % of ghost-rate labels saturate at exactly 0 or 1), which caps the attainable R^2; the surrogate residuals already approach this floor, so deeper networks buy little. Second, the closed-form effective diameter reproduces the evaluator exactly for 16 of the 18 combinations and remains approximate for two-scale designs (up to ≈ 2 % from the discrete checkerboard average) and for rings after integer rounding at decoding (up to ≈ 15 %), and surrogate error near the sharp ghost-onset boundary is not cosmetic: at $M = 1$ D, $p = 4$ mm the surrogate's top candidate (two-scale/ellipse) was predicted at 0.687 but verified at 0.598, an overestimate of 0.09 that only the mandatory top-three true re-evaluation caught; even the re-ranked winner of that condition carries the largest residual deviation (0.016) of the eight. Surrogate optimization must always end with true-model verification, especially near parameter boundaries. Third, the sigmoid projection confines the search to the sampled legal domain; extrapolation to larger pitches or new shapes is untested. Fourth, path B inherits its teacher ceiling (Sec. 4.3), and integer parameters (ring counts, aperture numbers) are optimized as continuous and rounded at decoding, a loss we did not observe but cannot exclude in general.

5 Results: optimized and fusion-compatible designs

5.1 Score-optimal designs of the closed-form model

Maximizing the closed-form score (11) (a 200×200 grid search with Nelder–Mead refinement performed on a neural surrogate of the closed-form score, validation $R^2 = 0.9994$, every optimum then re-evaluated with the exact model, maximum deviation 0.82 %) for $M \in \{1, 2, 3, 4\}$ D on square and hexagonal lattices yields the designs of Table 4; the score landscape for the representative case $M = 3$ D is shown in Fig. 6. Three robust features emerge. First, all optima converge to a narrow window, $d \approx 520$–553 µm, $s \approx 1.24$–1.29 mm and $\eta \approx 0.15$, the latter coinciding with the peak of the fill-factor membership. The optimal diameter decreases only weakly with M (553 → 520–545 µm from $M = 1$ to 4 D), much more weakly than the $M^{-1/2}$ scaling of the acuity-only criterion (5), because the score is co-determined by fill factor and coverage (Fig. 7). Second, square and hexagonal lattices perform almost identically (score differences below 0.3 %). Third, the optimal diameters are quantitatively consistent with the diffraction–defocus balance: Eq. (5) gives 546 µm at $M = 3$ D, within 4 % of the numerical optima, and roughly three times the classical zero-defocus criteria of Young (172 µm [1]), of Newman and Rible (209 µm, quoted in Ref. [2]) and of the OTF analysis of Ref. [5] (≈ 0.2 mm), because those criteria assume an emmetropic eye while the intended users are uncorrected myopes. Clinical data point the same way: 0.5–0.75 mm pinholes are diffraction-limited while 1 mm pinholes tolerate 5 D of induced error at the 20/40 level [4].

The most striking feature of these optima concerns the ghost term: for all eight designs the eye-rotation metric gives $g \approx 40$–$66 \gg 1$, and indeed $g \gg 1$ throughout the feasible region at $L = 15$ mm. The ghost membership therefore saturates near zero and contributes a nearly constant offset, so the closed-form optimization is driven by acuity, fill factor and coverage alone (Fig. 8). The normalized acuity of the optimized devices, $a \approx 0.16$–0.26, remains below the emmetropic diffraction limit by construction but corresponds to an order-of-magnitude reduction of the blur circle relative to the unaided myopic eye, for which $c_{\mathrm{geom}} = p\,M/P$ with the full pupil.

5.2 Fusion-compatible triple-win designs

Re-evaluating the score optima with the calibrated Scheiner metric (8) changes the ghost picture quantitatively. For the representative $M = 3$ D square-lattice optimum, g_{Sch} drops from 1.62 under the uncorrected criterion $\Delta x > c_{\mathrm{tot}}$ to 1.14 once the fusion floor is applied: the score-

Table 4: Score-optimal pinhole-sunglass designs for myopia $M = 1$–4 D on square and hexagonal lattices (closed-form score (11), exact-model values; coverage equals unity for all designs). The ghost column g is the eye-rotation metric (9) entering the score; under the calibrated Scheiner metric (8) the eight designs span $g_{\mathrm{Sch}} = 0.38$–1.54 (fused for $M \leq 2$ D, just outside the fusion regime for $M \geq 3$ D), analysed in Sec. 5.2.

M (D)	lattice	d (ţm)	s (ţm)	η	a	g	S
1	square	552	1266	0.149	0.261	65.7	1.487
1	hex	525	1294	0.149	0.251	64.4	1.484
2	square	553	1261	0.151	0.228	57.2	1.476
2	hex	526	1290	0.151	0.224	57.3	1.474
3	square	551	1252	0.152	0.193	48.0	1.464
3	hex	525	1281	0.152	0.193	49.0	1.464
4	square	545	1240	0.152	0.164	40.4	1.454
4	hex	520	1270	0.152	0.166	41.8	1.455

Table 5: Highest-acuity designs satisfying all three constraints of the calibrated ghost analysis (fusion $g_{\mathrm{Sch}} \leq 1$, full pupil coverage, $\eta \in [0.05, 0.25]$), square lattice, $p = 2$ mm, evaluated with the exact physical model. The s-window column gives the feasible pitch interval at the listed diameter.

M (D)	d (ţm)	s (ţm)	s window (ţm)	η	a	g_{Sch}	cov.
1	720	1279	1277–1279	0.249	0.311	0.388	1
2	648	1352	1149–1352	0.181	0.236	0.819	1
3	529	1099	939–1099	0.182	0.193	0.999	1
4	456	824	809–824	0.241	0.167	0.999	1

optimal designs sit just outside the fusion regime. The correction reopens a design window that the uncorrected criterion had closed, because the fusion condition relaxes to $\Delta x \leq c_{\min}$, i.e. $s \leq c_{\min} P/M$ (1.10 mm at $M = 3$ D), which now overlaps both the fill-factor window and the dead-zone condition $s \leq p - d$. Figure 9(a) shows the three constraints in the (d, s) plane for $M = 3$ D and $p = 2$ mm: a triple-win region with $g_{\mathrm{Sch}} \leq 1$, full pupil coverage and $\eta \in [0.05, 0.25]$ exists, spanning $s \approx 0.94$–1.10 mm at $d \approx 0.53$ mm.

Table 5 lists, for $M = 1$–4 D, the feasible design with the highest acuity (ties broken towards $\eta \approx 0.15$). Fusion costs essentially no acuity: at $M = 3$ D the triple-win design keeps $a = 0.193$, identical to the score optimum, and at $M = 1$ D the acuity even improves (0.311 versus 0.261) because the constraints push the diameter closer to the diffraction–defocus balance. What tightens with M is the width of the window, which at $M = 4$ D narrows to $s \approx 0.81$–0.82 mm at $d \approx 0.46$ mm. The feasible region shrinks rapidly with myopia [Fig. 9(b)]: from 0.275 mm^2 for $M \leq 2$ D to 0.068 mm^2 at $M = 4$ D and $\lesssim 0.04$ mm^2 for $M \geq 5$ D, where it survives only at small diameters with reduced acuity ($a \lesssim 0.15$). Dilating the pupil from 2 to 4 mm enlarges the region only for $M \lesssim 2$ D (to 0.936 mm^2 at $M \leq 1$ D), where the dead-zone constraint binds; for $M \geq 3$ D the fusion and fill-factor constraints bind first.

Two limitations of this analysis should be stressed. First, the metric (8) evaluates the nearest-neighbour separation only; a strict any-pair test at a nearest-neighbour-fused design still flags ghosts with probability one (on average 5.75 beams enter a 2 mm pupil simultaneously at such designs, and 55–67 % of the beam pairs remain resolvable for $d = 0.40$–0.53 mm), because widely separated pairs near the pupil rim exceed $c_{\min}$ even when nearest neighbours fuse. The triple-win designs therefore *soften* monocular ghosting rather than abolish it. Second, the fusion floor reproduces the threshold behaviour of Ref. [4] only qualitatively: it places the diplopia onset at $M^* \approx 1.1$ D, whereas the clinical value is ≈ 4 D; matching it quantitatively would require $c_{\min} \approx 200$ ţm (≈ 31 arcmin), which is perceptually implausible and indicates that

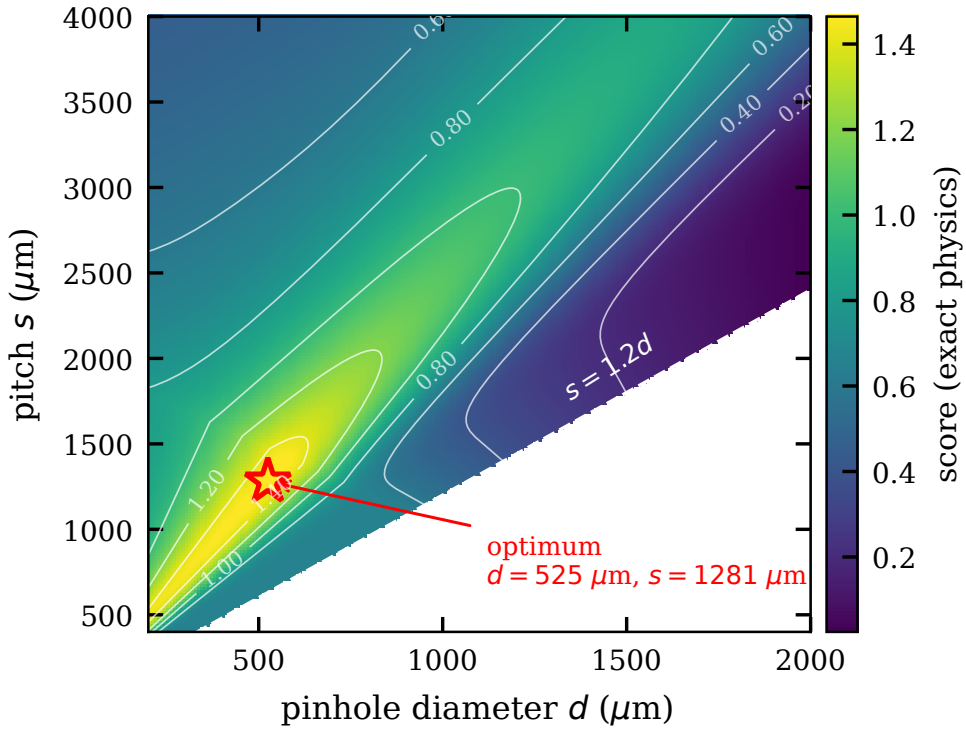


Figure 6: Composite score $S(d, s)$ evaluated with the exact physical model for $M = 3$ D on a hexagonal lattice. The dashed white line is the non-overlap boundary $s = 1.2\,d$ (the region below it is infeasible). The red star marks the optimum ($d = 525$ ţm, $s = 1281$ ţm).

higher-order perceptual fusion effects, beyond a geometric separation criterion, set the clinical threshold. Finally, eliminating ghosts entirely by enforcing single-beam entry ($s \geq p + d$) locks the blind-zone rate at its geometric minimum, $1 - \pi/4 \approx 21.5\,\%$ for a square lattice (the capture disc covers the fraction $\pi/4$ of the unit cell at $s = p + d$; hexagonal packing lowers this floor to $1 - \pi/(2\sqrt{3}) \approx 9\,\%$), which natural scanning eye movements cannot rescue (a 30-minute fixation–saccade simulation on a square lattice gives blind fractions of $\approx 21\,\%$ of the time); this topology-bound trade-off can be softened by denser packing but not eliminated.

6 Design specification

We condense the findings into a manufacturable specification. All numbers below are computed directly from the physical model of Sec. 2 (the generation script and its output are part of the code availability).

Universal main design. A single geometry serves the whole $M = 1$–4 D range: a hexagonal array of circular apertures with $d = 530$ ţm and pitch $s = 1.10$ mm at an eye relief $L = 15$ mm. The hexagonal lattice follows from the arrangement ranking of Sec. 3.1, the circular aperture from Sec. 3.2, and the pitch sits at the $M = 3$ D fusion edge $s = c_{\min} P/M$. The fill factor is $\eta = 0.211$ (hexagonal; the same d, s on a square lattice give 0.182), deliberately above the 0.15 membership peak because tightening the fill factor to 0.18 would require $s = 1.19$ mm and sacrifice fusion at $M = 3$ D ($g_{\mathrm{Sch}} = 1.08$). Table 6 gives the verified per-degree metrics: the design fuses nearest-neighbour images for $M \leq 3$ D (at $M = 3$ D exactly at the floor, $\Delta x = 55.0$ ţm$= c_{\min}$), shows mild resolvable ghosting at $M = 4$ D ($g_{\mathrm{Sch}} = 1.33$), and maintains full pupil coverage throughout. Degree-specific triple-win designs that also fuse at $M = 4$ D are listed in Table 5.

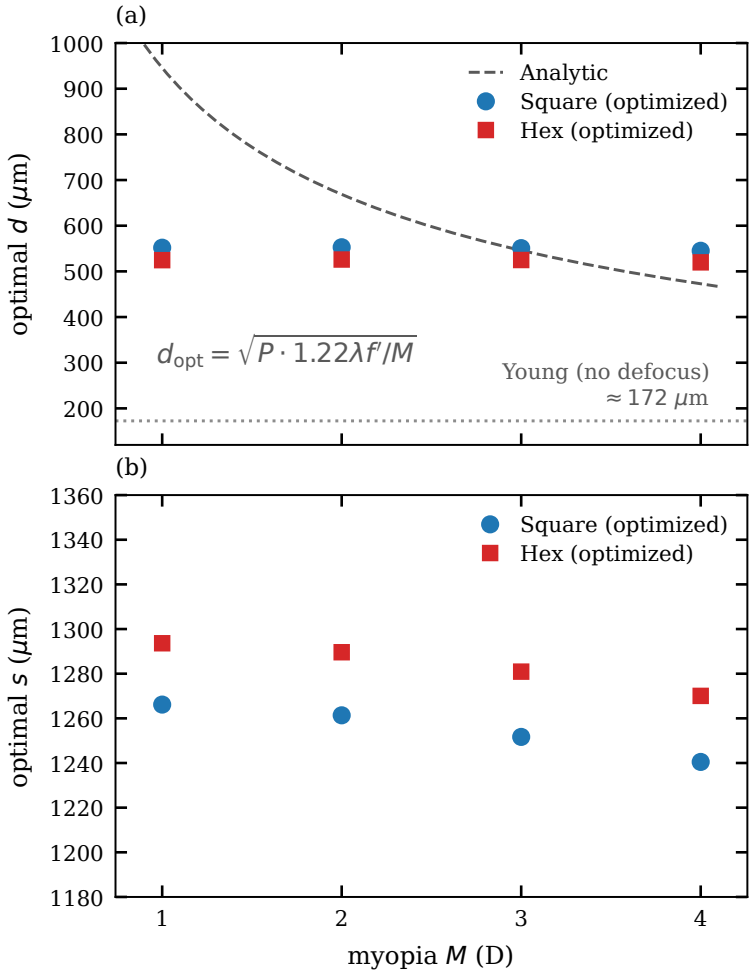


Figure 7: Optimized array geometry as a function of myopia M for square and hexagonal lattices (markers; values from Table 4). (a) Optimal pinhole diameter d; the dashed curve is the analytic diffraction–defocus balance of Eq. (5) and the dotted line the classical zero-defocus Young criterion $\sqrt{2.44\lambda f'} \approx 172$ μm. (b) Optimal pitch s. The optima depend only weakly on M and track the analytic balance, lying roughly three times above the Young value.

Film thickness and field of view. The geometrical half-angle of the field admitted by a single aperture of diameter d in a film of thickness t is $\arctan(d/2t)$. For the main design this gives 79.3° at $t = 50$ μm and 69.3° at $t = 100$ μm: because $d \gg t$, the film thickness barely restricts the per-aperture field, and standard 50–100 μm metal or blackened-polymer films are acceptable. The usable field is set by the lens aperture and eye relief, not by the film.

Transmission and sunglass classes. The total luminous transmittance is the product of the fill factor and the substrate transmittance T_{sub}. With $\eta = 0.211$, meeting the sunglass standard EN ISO 12312-1 requires $T_{\text{sub}} \geq 0.85$ for Category 2 (total transmittance 18–43 %) and $T_{\text{sub}} \in [0.38, 0.85]$ for Category 3 (8–18 %); the geometrical attenuation of the array alone is already a factor ≈ 4.7.

Aperture count. On a 40 mm diameter lens blank, the hexagonal unit cell of area $\sqrt{3}\, s^2/2 = 1.048$ mm^2 gives ≈ 1200 apertures.

Tolerances. A 9×9 grid scan over $d \pm 10$ % and $s \pm 5$ % around the main design (Table 7) shows that the acuity is robust (within ± 0.02 for all M, since the diffraction–defocus balance

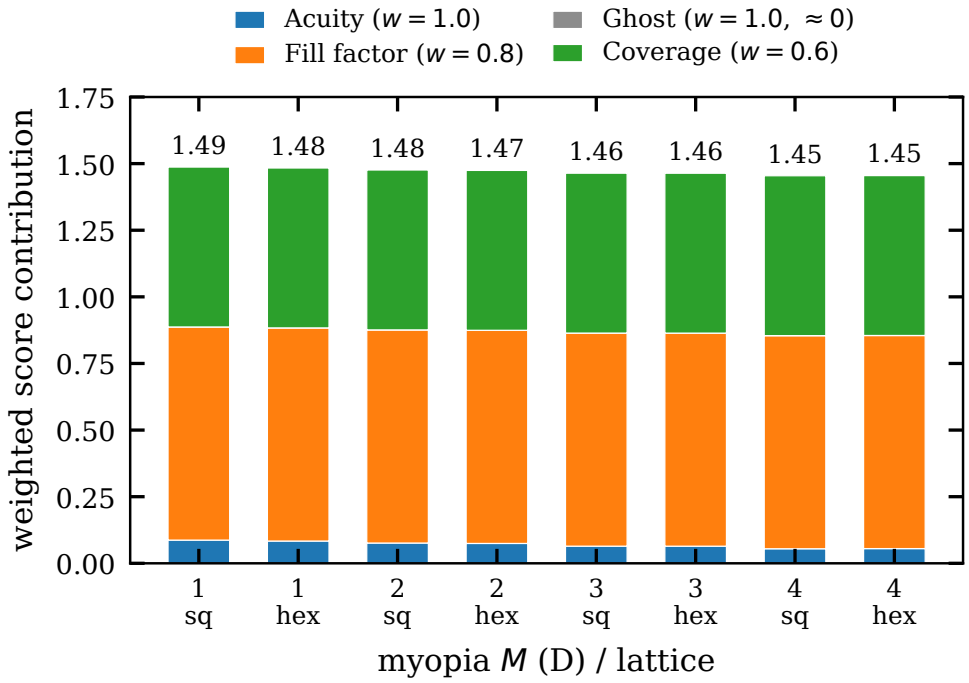


Figure 8: Decomposition of the composite score [Eq. (11)] into its four weighted contributions for the eight optimized designs of Table 4; bar labels give the total S. The ghost membership is pinned at zero for every design ($g \approx 40$–$66 \gg g_{\max} = 1$), so the ranking is decided entirely by the acuity, fill-factor and coverage terms.

Table 6: Universal main design (hexagonal lattice, circular apertures, $d = 530$ µm, $s = 1.10$ mm, $\eta = 0.211$, $L = 15$ mm): exact-model metrics per myopia degree at pupil $p = 2$ mm. Δx is the Scheiner separation (7); fusion requires $g_{\mathrm{Sch}} \leq 1$.

M (D)	a	c_{tot} (µm)	Δx (µm)	g_{Sch}	cov.
1	0.253	29.5	18.3	0.33	1
2	0.224	33.2	36.7	0.67	1
3	0.193	38.6	55.0	1.00	1
4	0.165	45.1	73.3	1.33	1

is second-order flat in d), the ghost metric scales linearly with s ($\pm 5\,\%$), and coverage stays at unity (the largest pitch, 1.155 mm, remains below $p - d \approx 1.42$ mm). The score's worst corner is driven by the fill factor: $d + 10\,\%$ with $s - 5\,\%$ pushes η to 0.282, where the Gaussian membership drops to ≈ 0.26; if score uniformity matters, the diameter tolerance should be tightened to $\pm 5\,\%$. The qualitative conclusions (fusion for $M \leq 2$ D across the whole tolerance band, the $M = 3$ D design astride the fusion edge, $M = 4$ D above it) are unchanged.

7 Discussion

Several limitations delimit the scope of these conclusions. First, the RSS combination of defocus and diffraction blur [Eq. (3)] is an engineering approximation: the classical treatments use separate geometric and diffractive regimes with a crossover criterion [1, 2], whereas a faithful treatment of a defocused pinhole in front of the eye requires the full defocused optical transfer function, as developed for single pinhole glasses by Abolhassani and Akbari [5]. Replacing Eq. (3) by an OTF-integral image-quality metric would be a natural high-accuracy upgrade; the surrogate and inverse-design framework would carry over unchanged. Second, the geometric blur

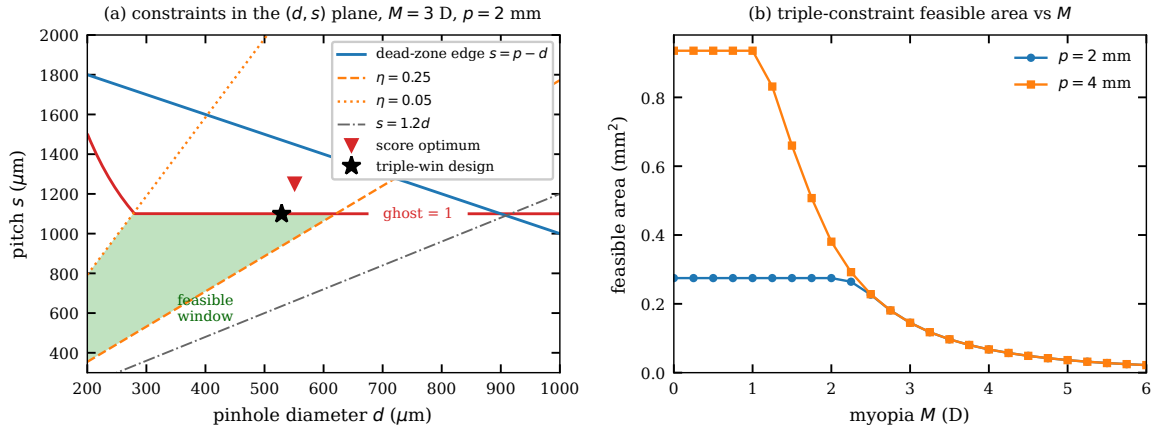


Figure 9: Ghost suppression with the calibrated Scheiner metric [Eq. (8)], square lattice. (a) Constraints in the (d, s) plane for $M = 3$ D and $p = 2$ mm: the fusion boundary $g_{\mathrm{Sch}} = 1$ (red; the upward turn at small d is the diffraction branch, where $c_{\mathrm{tot}} > c_{\mathrm{min}}$ widens the resolvability floor), the dead-zone edge $s = p - d$, the fill-factor curves $\eta = 0.25$ and 0.05 and the non-overlap line $s = 1.2d$ enclose the shaded triple-win region; the red triangle is the score optimum of Table 4 and the black star the highest-acuity triple-win design of Table 5. (b) Area of the triple-constraint feasible region versus myopia for pupil diameters $p = 2$ and 4 mm; pupil dilation enlarges the region only for $M \lesssim 2$ D.

Table 7: Tolerance analysis of the universal main design: metric ranges over a $d \pm 10\,\%$, $s \pm 5\,\%$ grid (9×9 points, exact model, hexagonal lattice, pupil 2 mm). Coverage equals 1.00 at every grid point.

M (D)	a range	g_{Sch} range	S range	η range
1	0.231–0.272	0.317–0.350	0.895–1.476	0.155–0.282
2	0.213–0.232	0.633–0.700	0.881–1.469	0.155–0.282
3	0.190–0.193	0.950–1.050	0.868–1.462	0.155–0.282
4	0.160–0.167	1.267–1.400	0.857–1.454	0.155–0.282

law $c_{\mathrm{geom}} = d\,M/P$ is the standard reduced-eye textbook relation; it is qualitatively supported by the scaling reported in Refs. [3, 4], but its coefficient 1/60 is not taken from the pinhole literature itself and could be refined with a more complete schematic eye. Third, Young's $\cos^4 \alpha$ illumination law [1] implies that peripheral pinholes transmit less effectively at large field angles, an effect not included in the fill factor of Eq. (6).

The ghost analysis carries its own caveats. The fusion floor $c_{\mathrm{min}} = 55$ ţm is a single calibration parameter anchored to one clinical geometry [4]; it reproduces the threshold behaviour only qualitatively (predicted onset ≈ 1.1 D versus the clinical ≈ 4 D), and the nearest-neighbour metric overstates the achievable fusion relative to a strict any-pair test (Sec. 5.2). The Monte-Carlo evaluator applies the any-pair test, but its shape treatment inherits the outer-diameter approximation, which is systematically optimistic for annular apertures because their sidelobe energy (up to 62.5 %) registers in no metric. The on-axis hexagonal ghost rate of zero in the directional test is an ideal-alignment result: real fixation wanders, so the practical advantage of hex over rings for central vision is real but smaller than the limiting values suggest. A direct experimental calibration of c_{min}, for example with two-pinhole masks of variable separation placed in front of a lens-defocused eye, would be the most valuable falsifiable test of the model.

The neural workflow also corrected an earlier methodological conclusion. For the closed-form model of Sec. 2, a surrogate adds interpolation error without speed benefit, and we previously advised against it; the Monte-Carlo evaluator, at 18–26 ms per evaluation, is expensive enough

that the surrogate route pays off after about 46 conditions (Sec. 4.4). The lesson is that the break-even point is set by the forward-model cost, and that surrogate optimization is only as trustworthy as its final true-model verification: the one substantial surrogate failure we observed (a 0.09 score overestimate for a two-scale design at the ghost-onset boundary) was caught by exactly this step. Within the training distribution the workflow is reliable; outside it, nothing is claimed. A second lesson concerns the objective: naively replacing the saturating ghost membership in Eq. (11) by its Scheiner counterpart makes the optimizer degenerate (it collapses to $d = 200$ µm, where the bloated diffraction blur formally "fuses" the images at a threefold acuity loss); a constrained formulation that keeps d at the diffraction–defocus balance avoids this failure mode.

Two findings point to perceptual experiments beyond the present metrics. The Vogel spiral's reduction of the pair-resolved fraction (diluting one strong double image into several weak ones) is an advantage that the ghost-rate-based score cannot reward; whether it translates into a measurable comfort benefit is an open psychophysical question. Likewise, the apodized taper's reduction of large-pitch blind zones at the price of transition-region ghosts is a trade-off whose perceptual optimum depends on how users weigh central clarity against field uniformity. Finally, we note the pedagogical value of the pipeline: the full chain, a transparent physical model, quasi-random sampling, Monte-Carlo evaluation, surrogate training, inverse design and exact-model validation, is compact enough for an advanced undergraduate project while illustrating a workflow that is now standard in computational physics and engineering design.

8 Conclusions

We have formulated the design of pinhole sunglasses as a quantitative multi-objective problem on a reduced-eye model with a calibrated Scheiner-disc ghost criterion, mapped the joint design space of aperture shapes and lattice arrangements under a fixed fill factor, and shown how neural surrogates accelerate the search. Circular apertures are optimal by the isoperimetric argument, with the highest acuity and lowest ghost rate among the genuinely non-circular shapes in all tested conditions; elliptical apertures cannot exploit their anisotropy and are quantitatively ruled out; annular apertures pay an unrecorded sidelobe penalty. Hexagonal packing leads the hard-condition arrangement ranking (best score 0.693), with the Vogel spiral reducing ghost severity rather than occurrence and ring lattices unsuitable for central-vision priority. The calibrated fusion floor ($c_{\min} \approx 55$ µm) opens a triple-win window: fusion-compatible designs with full pupil coverage exist for $M = 1$–4 D ($d = 456$–720 µm, $s = 824$–1352 µm, $\eta = 0.18$–0.25), and a universal hexagonal design ($d = 530$ µm, $s = 1.10$ mm, $\eta = 0.211$, about 1200 apertures per 40 mm lens) covers $M = 1$–3 D with fused nearest-neighbour images. A surrogate of the array evaluator ($R^2 = 0.979$–0.995) enables gradient-ascent inverse design in 3.7 s per condition (4.2× over direct search at statistically equal quality) and single-pass inverse design in 0.5 ms, provided every optimum is verified against the true model. The boundaries are explicit: fusion of nearest neighbours softens but does not abolish ghosting, the fusion floor is a one-parameter qualitative calibration, and the neural claims hold only within the sampled distribution.

Data availability

The code implementing the physical model, the Monte-Carlo array evaluator, dataset generation, surrogate and inverse-network training, the inverse-design benchmark and all figure-generation scripts, together with the datasets, the trained checkpoints and the specification-generation script, is available from the authors on reasonable request. The early closed-form optimization pipeline behind Table 4 was superseded during repository consolidation and is no longer part of the codebase; its designs are preserved in the repository (`results.csv`), and every value

reported in Sec. 5.1 is re-computed from the exact physical model by the surviving figure-generation scripts.

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